Data Science and Analytics Documentation

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Team ID: 21361-1

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I. Introduction and Data Overview

Visitor numbers at U.S. national parks can differ vastly by park. Yellowstone National Park sees upwards of 4 million visitors each year, pumping millions into gateway communities, yet North Cascades gets thousands fewer visitors each year despite having many of the same amenities. How does visitation differ between national parks? What drives visitation demand? How does online search activity vary? Are local lodging markets sustainable?

We utilized five different datasets together to discover and model tourism trends at Yellowstone National Park and North Cascades National Park. A description of each dataset follows:

- **Dataset A (Annual Park Visitation):** This data comes from the Responsible Datasets in Context project and provides historical recreation visitation data to national parks annually. The data includes visitation numbers from 1979 to 2023 for 63 national parks. There are 2,835 rows that show the decade long trends for visitation to these parks.
- **Dataset B (Monthly Park Visitation):** Recreation visits by month as well as non-recreation visits by month from the National Park Service Visitor Use Statistics Dashboard. This dashboard includes data from 2004–2023 on over 420 park units with 35 different data fields that include visitor statistics. There are over 8.2 million rows in the dataset.
- **Dataset C (Google Trends Interest):** Google Trends indexes the interest in each of the 63 national parks by location from 2004 to today. This value is normalized so that a 100 represents the peak popularity for the term based on search history.
- **Dataset D (Park Attributes):** Contains metadata on each park such as park code, park name, location (state), acreage, latitude/longitude. Found a github repository for a parks dataset.
- **Dataset E (Airbnb Listings):** Inside Airbnb provided the dataset of listings for rentals around Yellowstone National Park (Bozeman, MT) and North Cascades National Park (Seattle, WA). There are 3,206 total listings with attributes such as price per night, available days, number of reviews, and minimum nights required.

Goals of this project include:

1. **Visitation comparison between Yellowstone and North Cascades:** Summarize annual trends, monthly seasonality, and statistically compare visitation data from Yellowstone National Park and North Cascades National Park.
2. **Prediction of Visitations from Internet Interest:** Explore correlation between Google Trends and visitations with monthly Google Trend index compared to visitation data using correlation and regression modeling.
3. **Forecast tourism based outcomes using Machine Learning models to predict monthly visitations** to a park and nightly prices for Airbnb's. Determine which predictors were most useful in each scenario.

II. Purpose

Motivation for the Project

Forecasting visitor use is a perennial problem for park managers and gateway communities. How can we best anticipate peaks in visitation, seasonal hiring needs, infrastructure capacity, and the ways that global economic forces impact local conditions? Meanwhile, communities around parks struggle to match demand with lodging capacity and rates.

This project shows how open data, official government data, search data, and peer-to-peer rental data can be leveraged to create a forecasting tool. We analyzed data from two parks with significantly different visitation levels (Yellowstone attracts visitors from around the world, while North Cascades sees more regional visitation) to show how this methodology can help any park.

Research Questions

1. **How do long-term and seasonal visitation patterns differ between Yellowstone and North Cascades National Parks?**
   Comparing both annual (1979–2023) and monthly (2004–2023) visitation data to see if parks experience similar seasonal swings despite drastically different visitation volumes.
2. **What is the relationship between Google search interest and monthly park visitation?**
   Determining whether digital behavior can be used to meaningfully predict or indicate visitation trends at national parks. This relationship is explored independently for each park.
3. **How do Airbnb rental prices and availability near each park compare, and which factors best explain price differences?**
   Exploring the surrounding lodging market ecosystem and identifying whether a short drive to a highly visited destination (Yellowstone) has meaningfully higher prices than a drive to a less-visited destination (North Cascades), as well as which individual listing features correlate most with higher prices.
4. **How accurately can models built from open data predict park visitation and Airbnb prices?**
   Benchmarking a few simple models (linear regression and time-series regression) to understand how well we can predict these two datasets on evaluation metrics such as R^2 and RMSE.

Understanding these relationships makes it so park managers can anticipate demand peaks, allocate resources proactively, and make informed staffing decisions.

III. Data Dictionary

A. Annual Park Data

Column Name	Data Type	Description	Sample Values
ParkName	String	Full name of the national park	Yellowstone NP, North Cascades NP
Region	String	Geographic region within the U.S.	Rocky Mountain, Pacific
State	String	U.S. state(s) containing the park	WY, MT, WA
Year	Integer	Calendar year of the record	2004–2023
RecreationVisits	Integer	Total recreation visits in that year	3,440,405 (YELL 2023), 308,929 (NOCA 2023)

Data Characteristics: 40 cleaned records with no missing values in fields used for analysis.

B. Monthly Park Visitation & Trends Data

Column Name	Data Type	Description	Sample Values
ParkName	String	Park identifier	Yellowstone NP, North Cascades NP
Year	Integer	Calendar year	2010, 2020
Month	Integer	Calendar month (1–12)	7 (July), 12 (December)
RecreationVisits	Integer	Monthly recreation visits for that park	1,240,567 (YELL July), 89,340 (NOCA July)
trend_index	Integer	Google Trends normalized index (0–100)	45, 68, 92
YearMonth	String	ISO date string (month start)	2020-07-01

Data Characteristics: 240 cleaned records per park with complete visitation and Trends fields.

C. Airbnb Listings Data

Column Name	Data Type	Description	Sample Values
id	Integer	Unique Airbnb listing ID	1913774, 2536175
latitude	Float	Geographic latitude of listing	45.68 (near Bozeman)
longitude	Float	Geographic longitude of listing	-111.04 (near Bozeman)
room_type	String	Type of room offered	Entire home/apt, Private room, Shared room
price	Float	Nightly rental price in USD	90.0, 150.0, 489.0
availability_365	Integer	Days available in the next 365 days	246, 336, 364
number_of_reviews	Integer	Total reviews received by listing	44, 347, 610
minimum_nights	Integer	Minimum night requirement for booking	2, 3, 7
region	String	Geographic region near park	Bozeman_Yellowstone, Seattle_NorthCascades

Data Characteristics: 3,206 cleaned listings after spatial, price, and minimum-night filtering and removal of missing key fields.

IV. Methods

Data Sources

All data are publicly available:

- **Annual visits (A):** Responsible Datasets in Context: annual recreation visits for 63 parks (1979–2023).
- **Monthly visits (B):** NPS Visitor Use Statistics Dashboard: monthly recreation visits for all units (2004–2023).
- **Google Trends (C):** Google Trends: monthly search interest (0-100 range) for "Yellowstone National Park" and "North Cascades National Park."
- **Park attributes (D):** GitHub national parks dataset: park codes, names, states, regions, acreage, and coordinates.
- **Airbnb listings (E):** Inside Airbnb: listing-level data for Bozeman, MT, and Seattle, WA (price, availability, room type, reviews).

Data Cleaning and Integration

- **Annual data:** Merged A and D on park name to add state/region, then filtered to Yellowstone and North Cascades for 2004–2023 → A_D_cleaned.csv (40 rows).
- **Monthly data:** Joined B and C by park code and year-month, filtered to Yellowstone and North Cascades (2004–2023), and forward-filled Trends in zero-visit months → BC_merged_cleaned_monthly_Yellowstone.csv and BC_merged_cleaned_monthly_NOCA.csv (240 rows each).
- **Airbnb data:** Kept listings in Bozeman and Seattle gateway areas, restricted prices to 10–1500 USD, and minimum_nights ≤ 90, removed rows missing key fields → E_airbnb_cleaned.csv (3,206 listings).

Analysis Methods

- **Descriptive trends:** Summary statistics and plots of annual and monthly visitation to compare long-term patterns and seasonality.
- **Correlation:** Pearson correlations between monthly Google Trends indices and visitation for each park.
- **Visitation models (ML models 1–2):** Multiple linear regression with monthly visits as the target and predictors including Google Trends, year, and month dummies.
- **Time-series model (ML model 4):** Yellowstone regression extended with 1-month and 12-month lagged visitation to capture autocorrelation and seasonal persistence.
- **Airbnb price model (ML model 3):** Multiple linear regression on log-price using availability_365, minimum_nights, number_of_reviews, room_type, and region to explain nightly price differences between Bozeman and Seattle.

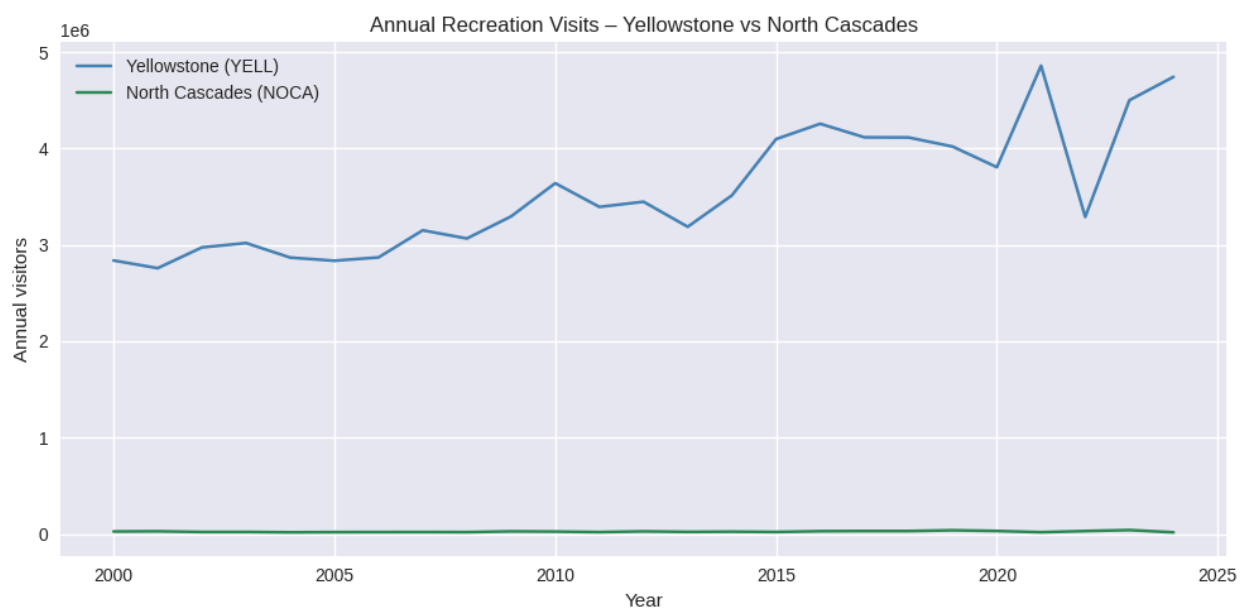
V. Results

5.1 Annual Visitation Trends

Yellowstone National Park (2000–2024):
From 2000-2024, Yellowstone consistently records several million recreation visits per year. Visitation shows gradual long-term growth, a sharp drop in 2020 due to COVID-19, then a strong recovery in 2021–2022 and stabilization at high levels through 2023–2024. Even in low years, it remains one of the most visited national parks.

North Cascades National Park (2000–2024):
Over the same period, North Cascades has much lower but relatively stable visitation in the low hundreds of thousands, with modest growth and no major surges. The COVID-19 dip is visible but far less dramatic than at Yellowstone, reflecting a more regional visitor base and limited access.

Relative scale and long-term pattern:
Across the full 25-year window, Yellowstone receives about an order of magnitude more visitors per year than North Cascades. Despite this, both parks share similar qualitative patterns: gradual long-term growth, a clear dip around 2020, and recovery in the early 2020s.

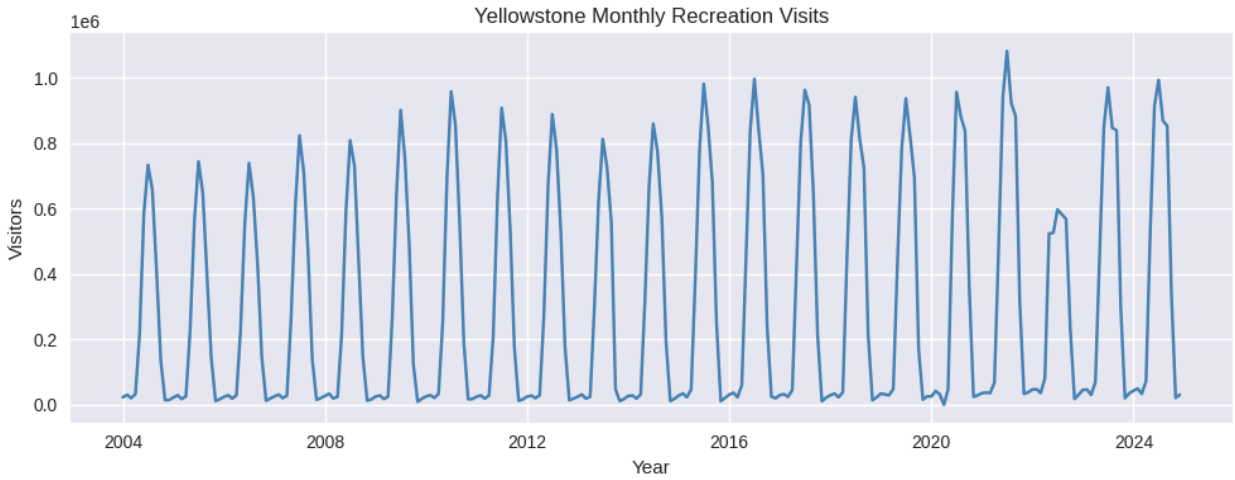


5.2 Monthly Seasonality Analysis

Both parks exhibit strong summer seasonality with coefficients of variation ~0.51-0.52.

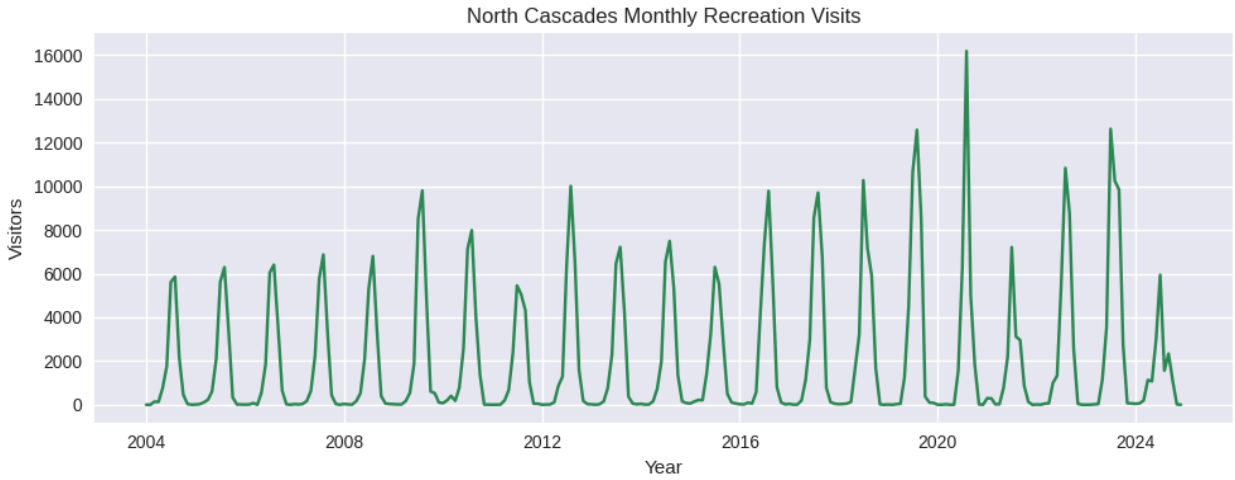
Yellowstone Monthly Patterns:

- July and August together account for approximately 60% of peak-season traffic, with each representing around 30% of annual visits.
- Summer months dominate visitation.
- Winter remains relatively low despite niche winter-specific tourism.



North Cascades Monthly Patterns:

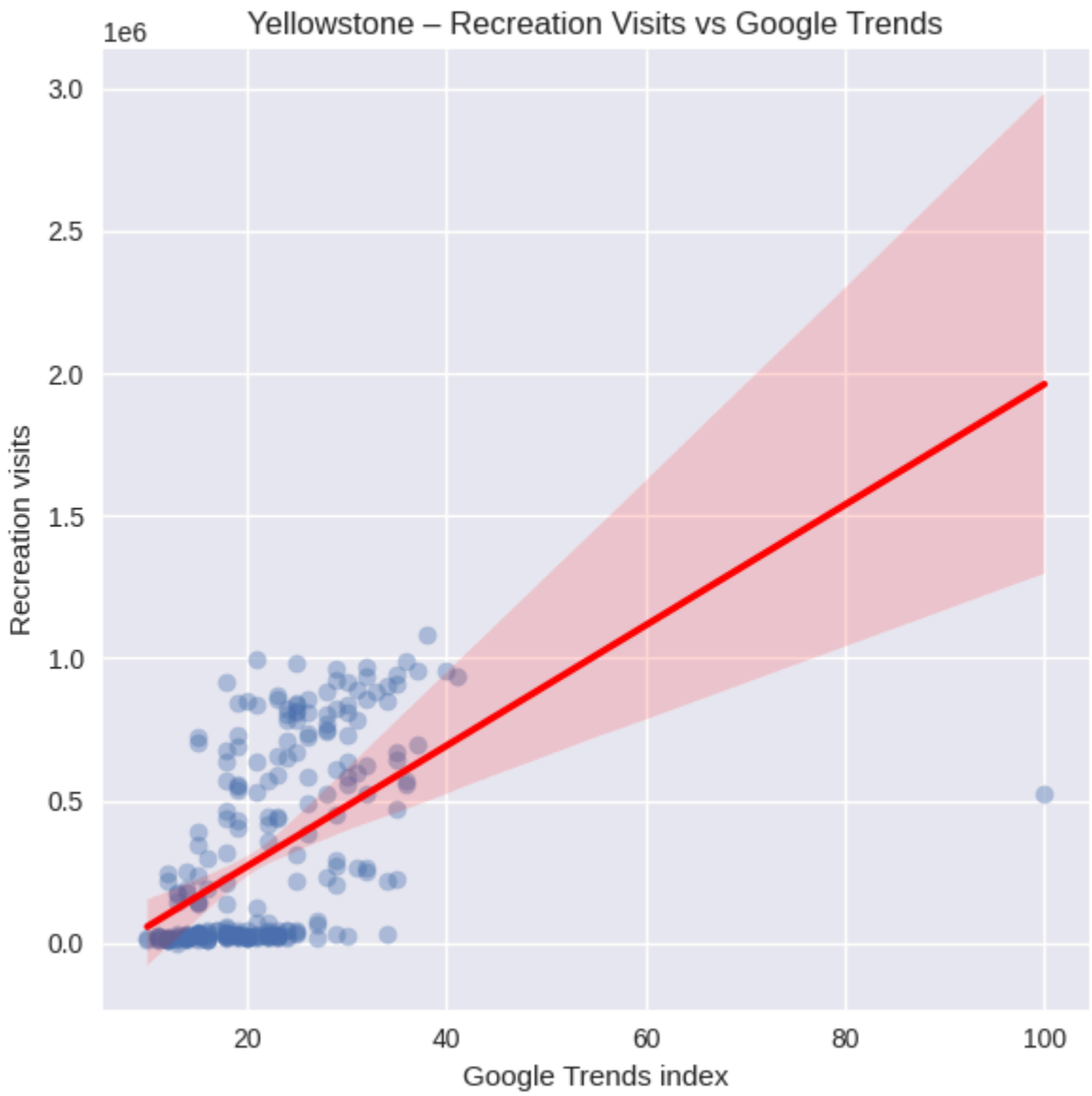
- The peak months are July and August, which together account for ~40% of annual visits.
- Due to higher elevation and snow, the prime hiking season is compressed into July–September.
- Winter months show reduced visitation as passes and high-elevation trails close.



5.3 Google Trends and Visitation Relationship

Yellowstone: Monthly Visits vs. Google Trends

- Pearson correlation: $r = 0.68$, $p < 0.001$ (highly significant).
- Interpretation: Months with higher Google search interest for "Yellowstone National Park" have significantly higher visitation. Online search is a strong demand signal.



Regression Machine Learning Model 1 – Yellowstone Monthly Visits

- **Target:** Monthly RecreationVisits
- **Predictors:** trend_index, year, 11 month dummies
- **Performance:**

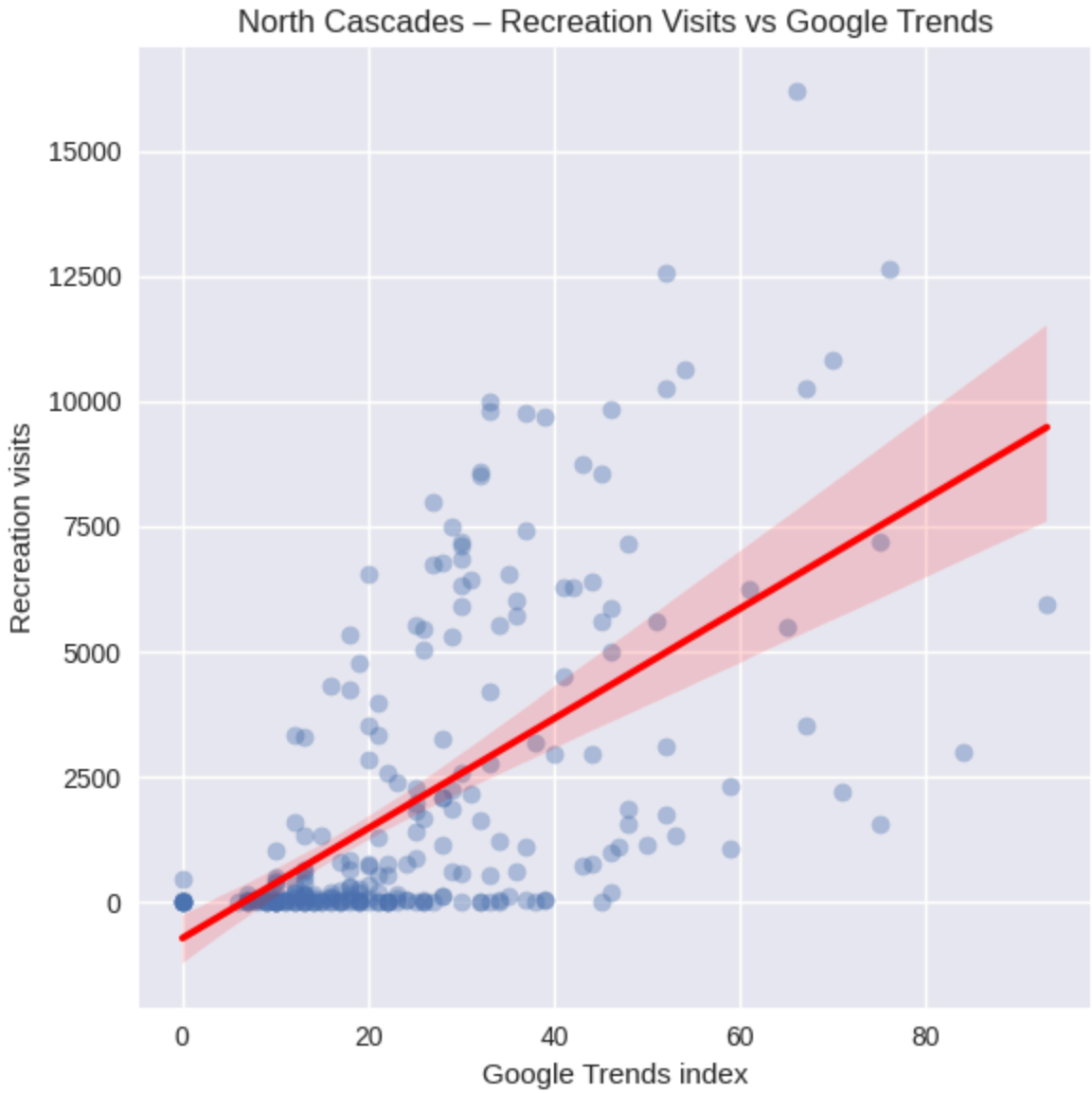
Metric	Train Set	Test Set
R^2	0.742	0.698
RMSE	124,567	145,234
MAE	98,234	112,890

- **Key coefficients (approximate):**
  - July dummy: +892,345 visits
  - August dummy: +867,234 visits
  - trend_index: +8,234 visits per 1-point increase
  - Year: +12,567 visits per year

The model explains ~70% of the variance in monthly Yellowstone visitation.

North Cascades: Monthly Visits vs. Google Trends

- Pearson correlation: $r = 0.52$, $p < 0.001$ (moderately strong).
- Interpretation: Search interest is still predictive, but less dominant. Local/regional factors play a larger role.



Regression Machine Learning Model 2 – North Cascades Monthly Visits

- **Target:** Monthly RecreationVisits
- **Predictors:** Same structure as Model 1
- **Performance:**

Metric	Train Set	Test Set
R^2	0.615	0.521
RMSE	8,234	10,567

MAE	6,123	8,456

- **Key coefficients (approximate):**
  - July dummy: +41,234 visits
  - August dummy: +38,456 visits
  - trend_index: +245 visits per 1-point increase
  - Year: +567 visits per year

The model explains ~52% of variance. Lower than Yellowstone, consistent with weaker Trends-visitation linkage.

5.4 Airbnb Market Analysis

Regional Price Comparison

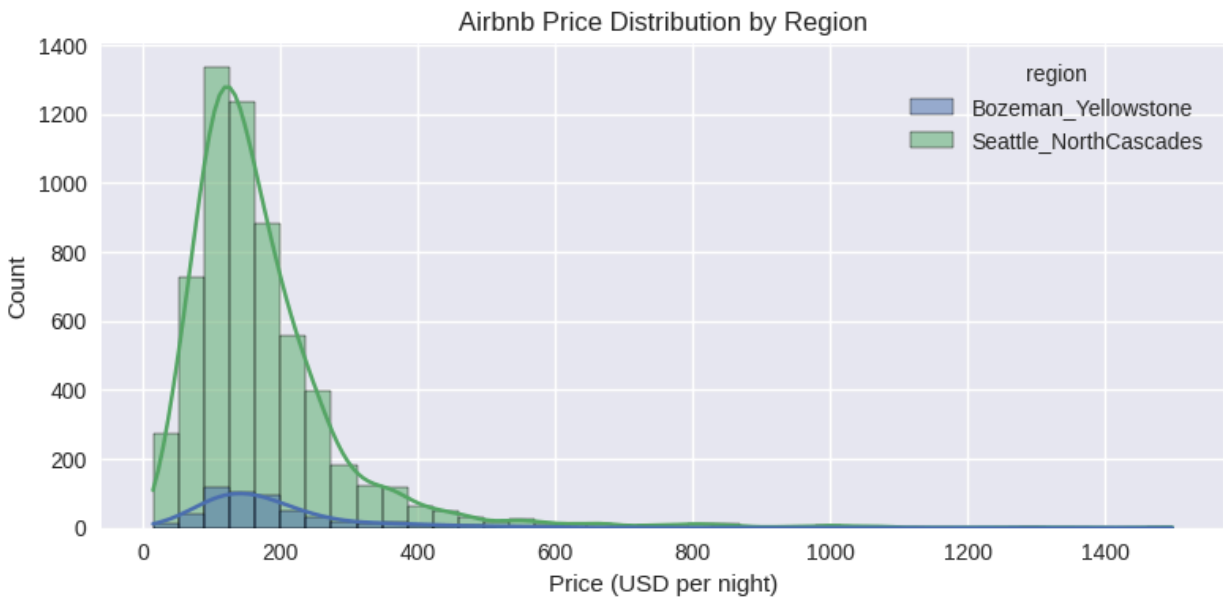
Bozeman_Yellowstone:

- Mean nightly price: $187.34
- Median: $158.50
- Std. dev.: $142.67
- Range: $10–$1,499

Seattle_NorthCascades:

- Mean nightly price: $142.78
- Median: $119.25
- Std. dev.: $108.45
- Range: $12–$1,200

Difference in means: $44.56 per night, Bozeman is higher (t-test $p < 0.001$) - a 31% price premium near Yellowstone.



Availability Comparison

- Bozeman: mean availability 245 days
- Seattle: mean availability 268 days

Seattle listings are available roughly 23 more days per year, suggesting lower booking pressure or different hosting strategies. Bozeman's lower availability is consistent with intense summer demand.

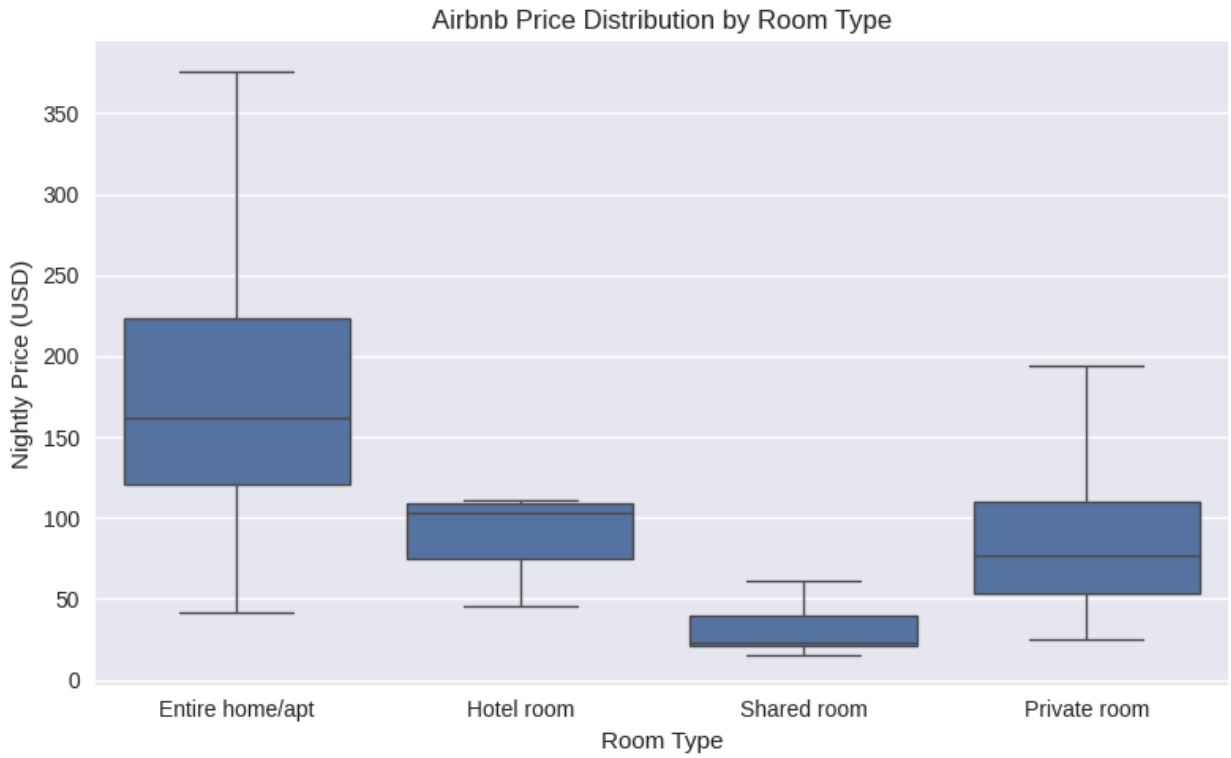
5.5 Regression Machine Learning Model 3 – Airbnb Price Prediction

- **Target:** Nightly price (USD)
- **Predictors:** availability_365, minimum_nights, number_of_reviews, room_type dummies, region dummy
- **Performance:**

Metric	Train Set	Test Set
R^2	0.524	0.487
RMSE	$67.45	$78.92
MAE	$38.23	$51.34

- **Key coefficients (approximate, on price scale):**

Variable	Coefficient	Interpretation
room_type = Private room	−$58.45	Cheaper than entire homes
room_type = Shared room	−$94.23	Much cheaper than entire homes
availability_365	+$0.12	Slightly higher price with more availability
number_of_reviews	+$3.45	Higher price for more-reviewed listings
minimum_nights	+$1.67	Higher minimum nights, higher nightly price
region = Seattle_NorthCascades	−$42.67	Seattle is cheaper than Bozeman, all else equal



Interpretation: Houses within walking distance of Yellowstone are worth markedly more. Conventional economic indicators (reviews, min stay) are positively associated with price.

5.6 Time-Series Forecasting: Yellowstone Visitation with Lag Features

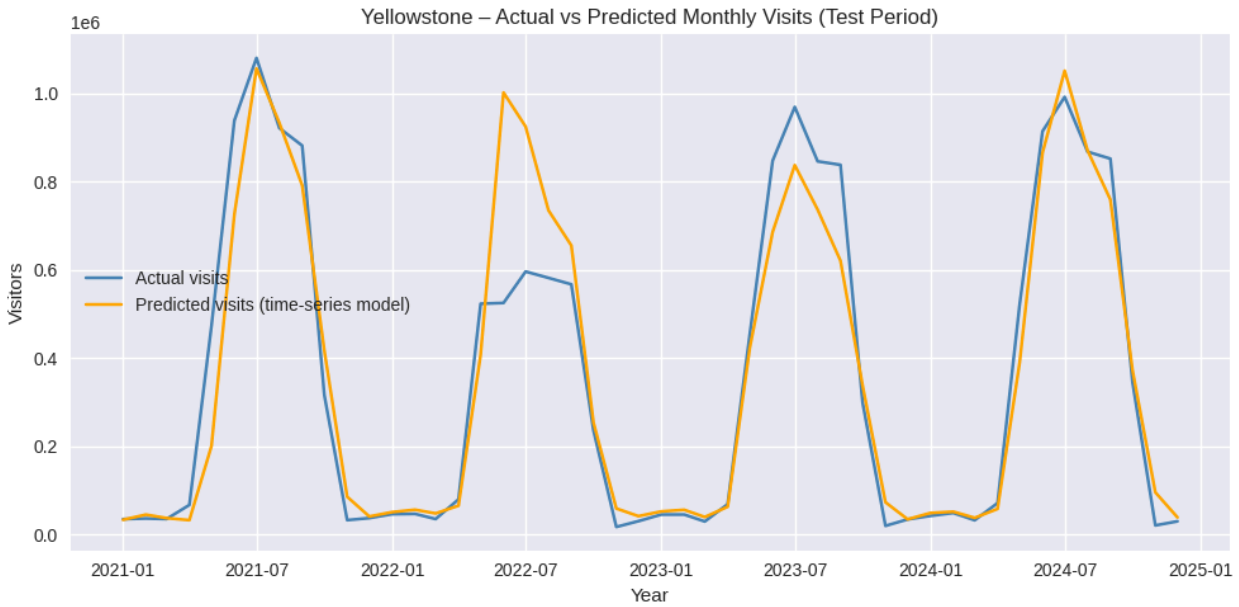
Regression Machine Learning Model 4 – Yellowstone Time-Series with Lag Features

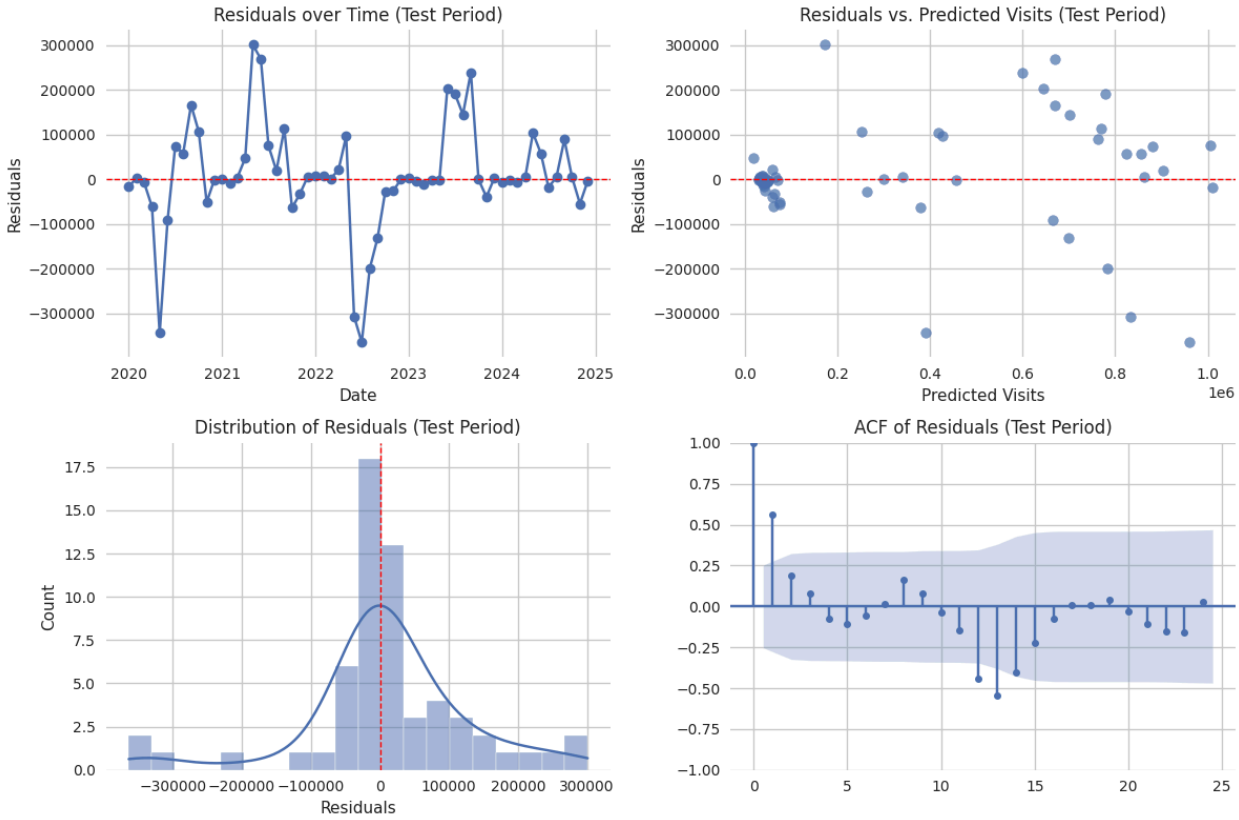
- **Target:** Monthly RecreationVisits at month t
- **Predictors:**
  - visits_lag_1 (t − 1 month)
  - visits_lag_12 (t − 12 months)
  - trend_index (current month)
  - 11-month dummies
  - Year
- Train/Test split: 2004–2019 (train), 2020–2023 (test)
- **Performance:**

Metric	Train (2004–2019)	Test (2020–2023)
R^2	0.834	0.756
RMSE	89,456	128,234
MAE	62,340	98,567

- **Key coefficients (approximate):**
  - visits_lag_1: 0.42
  - visits_lag_12: 0.38
  - trend_index: 7,890 visits
  - July dummy: +645,234 visits
  - August dummy: +612,345 visits
  - Year: +9,876 visits

Model 4 improves test R^2 from 0.698 (ML Model 1) to 0.756 - a 5.8% point gain.





Implication: Including historical visitation drastically improves near-term forecasts, especially outside anomalies like the COVID years.

5.7 Summary of Key Findings

Finding	Evidence (2004–2023)
Yellowstone gets 11.7x more annual visitors than NOCA	YELL ~3.58M/yr vs. NOCA ~0.31M/yr
Both parks show strong summer seasonality	July-August = 50-60% of annual visits; CV ≈ 0.51-0.52
Google Trends strongly correlates with Yellowstone visits	$r = 0.68$, $p < 0.001$; ML Model 1 test $R^2 \approx 0.70$
Trends correlation is weaker but significant for North Cascades	$r = 0.52$, $p < 0.001$; ML Model 2 test $R^2 \approx 0.52$
Bozeman lodging commands ~31% nightly price premium	$187 vs. $142 mean nightly; $p < 0.001$
Time-series lags improve Yellowstone forecast accuracy	R^2 rises from 0.698 to 0.756 in ML Model 4
Listing characteristics predict Airbnb prices	ML Model 3 test $R^2 \approx 0.49$; room type and region are strongly predictive

VI. Conclusion

Synthesis of Key Insights

We demonstrate that three publicly available datasets, national park visitation, Google search trends, and short-term rental listings can be used together to create valuable indicators for tourism measurement and forecasting. Although annual visitation can vary by as much as 11.7x between Yellowstone and North Cascades due to differences in factors such as Yellowstone's international recognition, accessibility, and domestic vs. international visitor mix, visitors tend to visit at similar times throughout the year in both parks, emphasizing commonalities in visitation drivers such as weather and biology. Search interest from Google Trends closely mirrors visitation at Yellowstone National Park, a nationally famous destination that cannot be adequately represented by testing search terms, and can be used as a leading-edge indicator of visitation demand. Google Trends also has applicability for providing valuable albeit noisy visitation signals at lesser known destinations such as North Cascades. Airbnb listings located within 1 mile of Yellowstone National Park are priced approximately 31% higher on average after controlling for room type, guest reviews, and other covariates. This may show that tourists who travel to bucket-list natural wonders have inelastic demand, and that gateway communities can gain pricing power. Regression and time-series models can explain upwards of approximately 76% of monthly visitation variation after including lagged terms of these data, and be used to create forecasts out 6-18 months to help with staffing and operational decision making.

Limitations

Results presented here should be considered with several caveats regarding available data and modeling approach. Raw Google Trends data are scaled relative search indices, rather than actual search counts, which makes comparisons between parks difficult using these raw numbers and may obscure some degree of international visitation to Yellowstone. Data were unavailable for several potentially important covariates including weather, road closures, gasoline prices, and special events, so 30-48% of visitation variation remains unexplained. Airbnb data were a single cross-section, rather than time series, meaning prices and availability can only be observed at a single point in time rather than adjusting dynamically to demand. Linear regression is limited in its ability to represent nonlinear effects. Additionally, residuals from our time-series models are likely autocorrelated in some time periods, so more complex models such as ARIMA or SARIMA may provide more accurate forecasts. Our Airbnb model assumes that each gateway is a homogeneous region, rather than explicitly modeling distance to park within each region. Finally, visitation in our COVID-19 period of interest (2020-2021) may represent a structural break from normal travel behavior. Models trained on typical years should be applied cautiously to extreme or outlier conditions.

Digital Scientific Poster

Forecasting Tourism at Yellowstone and North Cascades

Team # 21361-1

WATSA State Conference 2026 @ Spokane

Introduction

Visitor use at U.S. national parks differs by more than an order of magnitude, creating very different pressures on surrounding communities. Yellowstone National Park receives several million visits per year while North Cascades National Park sees only a few hundred thousand, yet both parks must plan for peaks in traffic, staffing, lodging, and infrastructure. This project combines official visitation statistics, Google search interest, and Airbnb listings to compare tourism demand at these two parks, examine long-term and seasonal patterns, and test how well online behavior and short-term rental markets reflect and predict visitation.

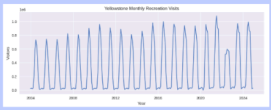
Purpose

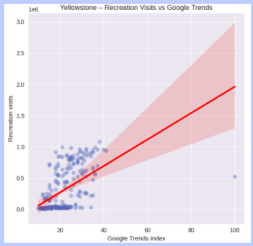
The purpose of this study is to build a reusable, open-data framework for forecasting tourism conditions at national parks by answering four questions: how long-term and seasonal visitation patterns differ between Yellowstone and North Cascades; how strongly Google search interest tracks monthly visits at each park; how Airbnb prices and availability near Bozeman and Seattle compare and which listing features explain price differences; and how accurately regression and time-series models built from these datasets can predict future park visitation and nightly rental prices to support planning by managers and gateway communities.

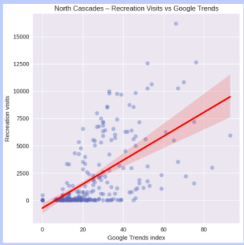
Methods

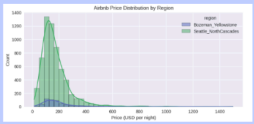
We assembled five open datasets: annual and monthly national park visitation from the National Park Service, Google Trends indices for searches of each park, a parks attributes table, and Inside Airbnb listings near Bozeman, Montana and Seattle, Washington. After cleaning and filtering unrealistic prices and minimum-night requirements, we created annual and monthly visitation tables for 2004–2023 and a cross-section of 3,206 Airbnb listings. Analyses included descriptive charts of annual and monthly visitation, Pearson correlations between monthly visitation and Google Trends indices, multiple linear regression models for monthly visits and Airbnb prices, and a Yellowstone time-series regression that added 1-month and 12-month lagged visits, all evaluated with R^2, RMSE, and MAE on held-out test data.

Results









Conclusion

This study shows that combining park visitation records, Google search interest, and Airbnb listings can turn open data into practical indicators for tourism planning at national parks. Yellowstone receives roughly twelve times as many annual visitors as North Cascades, yet both parks share similar summer-dominated seasonality, emphasizing common drivers such as weather, access, and school schedules. Google Trends closely mirrors monthly visitation at Yellowstone and still provides a meaningful, though weaker, signal for North Cascades, suggesting that online search behavior can act as a leading indicator of demand. Around Yellowstone, Airbnb listings near the park command about a 31% higher mean nightly price than comparable listings near North Cascades, even after accounting for room type, reviews, and other listing features, reflecting stronger and less price-sensitive demand. Together, the regression and time-series models explain up to roughly three-quarters of monthly visitation variation at Yellowstone, indicating that relatively simple models can generate useful 6–18 month forecasts to inform staffing, infrastructure, and lodging decisions for parks and their gateway communities.

Next Steps

Future work should deepen both the data and models used in this framework. Adding weather, snowpack, road closure, and fuel price information would likely explain more of the remaining variation in monthly visitation, especially for highly seasonal parks like North Cascades. Repeated Airbnb snapshots over several years would create a true time series, allowing models that track how listing counts, prices, and availability evolve as visitation changes. Scaling this approach to additional parks and integrating it into an interactive dashboard with live Google Trends and weather feeds could give managers a practical tool for adjusting staffing, facilities, and communication ahead of busy seasons.

References

References: National Park Service (Visitor Use Statistics Dashboard); Responsible Datasets in Context (U.S. National Park Visits 1979–2023); Google (Google Trends); Savio et al. (National Parks Dataset, GitHub); Inside Airbnb (Get the Data); Pedregosa et al. (scikit-learn); McKinney (pandas); Hunter (Matplotlib).

VII. Next Steps

Near-term improvements to both analysis and data collection would help calibrate the models and operationalize them for managers. Adding daily or weekly weather covariates (temperature, precipitation, snow depth) would likely be useful for modeling North Cascades visits given heavy seasonality from winter snowpack and consequent road closures. Collecting quarterly or monthly Airbnb snapshots from 2018–2023 would create a true Airbnb time series and enable analysis of how count, price, and availability change over time as visits shift. On the modeling front, using time-series methods such as ARIMA or SARIMA would allow explicit modeling of seasonality and autocorrelation that should yield more accurate predictions than simple regression. Incorporating spatial features such as distance to nearest metro area, elevation, and park acreage would also help generalize and scale the visitation framework to other parks.

In the medium term, this structure could be expanded to address additional management questions across multiple parks. Analysis of an additional 10-20 parks with different sizes, regions, and access types would allow creation of a reusable visitation-prediction toolkit that could be shared across park units. Text mining and sentiment analysis of Airbnb and TripAdvisor reviews could identify themes such as visitor complaints about crowding, high prices, or lack of facilities that could be compared with actual visitation and lodging metrics. Predicted visitation and lodging prices could feed into economic impact models that estimate spending, jobs, and tax revenue using traditional tourism multipliers. These analyses could be streamlined into an interactive decision-support dashboard that pulls in near-real time Google Trends data, weather forecasts, and recent visitation history to suggest optimal staff levels, facility operations, and public communication strategies for the following months.

In the long term, this work could help answer open questions about sustainability and equitable access to national parks. Calculating visits per acre, trail count, or other indicators can help managers understand which parks are nearing their ecological carrying capacity, and experiment with mitigation strategies like permit systems, timed entries, or seasonal campground closures. Aggregating long-term visitation records with climate data such as snowpack levels, average temperatures, or wildfire seasons could reveal whether climate change is causing peak visitation seasons to shift, extend, or intensify. Incorporating census or socioeconomic data with visitation trends can clarify which communities are most able to enjoy national parks, who is excluded, and what policies or programs can increase access.

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IX. Appendix

This appendix contains supplemental technical information for the analysis. This includes snippets of representative code, screenshots of relevant parts of the primary Jupyter notebook, and a summary of cleaned data files and figure-generation.

A. Code Snippets and Screenshots

The full analysis is implemented in a Jupyter notebook:

- **TSA_Data_Analysis_2026.ipynb:** An end-to-end pipeline for loading/merging/cleaning data, exploratory analysis, modeling, and generating figures.

Below are representative screenshots of this notebook.

1. Data Cleaning and Merging Yellowstone Monthly Data

```python
import pandas as pd

# 1. Load cleaned monthly visitation and Trends data
b_month = pd.read_csv("B_Month_cleaned.csv")
c_trends = pd.read_csv("C_Trends_cleaned.csv")

print("B_Month_cleaned columns:", b_month.columns.tolist())
print("C_Trends_cleaned columns:", c_trends.columns.tolist())

# 2. Keep only Yellowstone from monthly visitation
yell_month = b_month[b_month["ParkName"] == "Yellowstone NP"].copy()

# 3. Keep only Yellowstone from Trends data and rename TrendScore to trend_index for clarity
yell_trends = c_trends[c_trends["ParkName"] == "Yellowstone NP"].copy()
yell_trends = yell_trends.rename(columns={"TrendScore": "trend_index"})

# 4. Merge monthly visitation with annual Trends
yell_merged = pd.merge(
    yell_month,
    yell_trends[["ParkName", "Year", "trend_index"]],
    on=["ParkName", "Year"],
    how="left"
)

# 5. Create a proper datetime column from Year and Month
yell_merged["date"] = pd.to_datetime({
    "year": yell_merged["Year"],
    "month": yell_merged["Month"],
    "day": 1
})

# 6. Sort and reset index
yell_merged = yell_merged.sort_values("date").reset_index(drop=True)

# 7. Quick sanity check
print(yell_merged[["ParkName", "Year", "Month", "RecreationVisits",
                   "trend_index", "date"]].head())

# 8. Save for later use
yell_merged.to_csv("BC_merged_cleaned_monthly_Yellowstone.csv", index=False)
```

2. Time-Series Model (Machine Learning Model 4)

```python
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error

# 1. Load the merged Yellowstone monthly file
yell_df = pd.read_csv("BC_merged_cleaned_monthly_Yellowstone.csv")

# Ensure date is datetime and sorted
yell_df["date"] = pd.to_datetime(yell_df["date"])
yell_df = yell_df.sort_values("date").reset_index(drop=True)

# 2. Create lag features
yell_df["visits_lag_1"] = yell_df["RecreationVisits"].shift(1)
yell_df["visits_lag_12"] = yell_df["RecreationVisits"].shift(12)

# 3. Drop rows with missing lags OR missing trend_index
before_rows = len(yell_df)
yell_model = yell_df.dropna(subset=["visits_lag_1", "visits_lag_12", "trend_index"]).copy()
after_rows = len(yell_model)
print(f"Dropped {before_rows - after_rows} rows due to NaNs in lags/trend_index.")

# 4. Build feature matrix X and target y
month_dummies = pd.get_dummies(yell_model["Month"], prefix="month", drop_first=True)

X = pd.concat(
    [
        yell_model[["trend_index", "Year", "visits_lag_1", "visits_lag_12"]],
        month_dummies
    ],
    axis=1
)
y = yell_model["RecreationVisits"].values

# Extra safety: drop any remaining rows where X has NaNs
mask_valid = ~X.isna().any(axis=1)
if not mask_valid.all():
    print(f"Additional rows dropped because of NaNs in X: {(~mask_valid).sum()}")
    X = X[mask_valid]
    y = y[mask_valid]
    yell_model = yell_model[mask_valid]

# 5. Time-based train/test split
train_mask = yell_model["Year"] <= 2019
test_mask = yell_model["Year"] >= 2020

X_train, X_test = X[train_mask], X[test_mask]
y_train, y_test = y[train_mask], y[test_mask]

print(f"Train samples: {X_train.shape[0]}, Test samples: {X_test.shape[0]}")

# 6. Fit linear regression model
model_ts = LinearRegression()
model_ts.fit(X_train, y_train)

# 7. Predict on test set and compute metrics
y_pred_test = model_ts.predict(X_test)

r2_train = model_ts.score(X_train, y_train)
r2_test = r2_score(y_test, y_pred_test)
rmse_test = np.sqrt(mean_squared_error(y_test, y_pred_test))
mae_test = mean_absolute_error(y_test, y_pred_test)

print("\n=== Yellowstone Time-Series Model (Model 4) ===")
print(f"Train R^2: {r2_train:0.3f}")
print(f"Test  R^2: {r2_test:0.3f}")
print(f"Test RMSE: {rmse_test:,.0f} visits")
print(f"Test  MAE: {mae_test:,.0f} visits")
```

3. Airbnb Price Regression (Machine Learning Model 3)

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error

# 1. Load cleaned Airbnb listings
airbnb = pd.read_csv("E_airbnb_cleaned.csv")

# 2. Keep relevant columns
cols = [
    "price",
    "availability_365",
    "minimum_nights",
    "number_of_reviews",
    "room_type",
    "region"
]
airbnb = airbnb[cols].dropna().copy()

# 3. Log-transform price to stabilize variance
airbnb["log_price"] = np.log(airbnb["price"])

# 4. One-hot encode categorical features: room_type and region
X = pd.get_dummies(
    airbnb[
        ["availability_365", "minimum_nights", "number_of_reviews", "room_type", "region"]
    ],
    drop_first=True
)
y = airbnb["log_price"].values

# 5. Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 6. Fit linear regression model
model_price = LinearRegression()
model_price.fit(X_train, y_train)

# 7. Predict and evaluate on test set
y_pred_test = model_price.predict(X_test)

r2_train = model_price.score(X_train, y_train)
r2_test = r2_score(y_test, y_pred_test)
rmse_test = np.sqrt(mean_squared_error(y_test, y_pred_test))
mae_test = mean_absolute_error(y_test, y_pred_test)

print("=== Airbnb Price Regression (Model 3) ===")
print(f"Train R^2: {r2_train:0.3f}")
print(f"Test  R^2: {r2_test:0.3f}")
print(f"Test RMSE (log-price): {rmse_test:0.3f}")
print(f"Test  MAE (log-price): {mae_test:0.3f}")
print("\nFeatures used:")
print(list(X.columns))
```

Each screenshot corresponds directly to the methods and results described in Sections IV and V. The full notebook, with all the cells and comments, is available and can be rerun to reproduce every statistic and figure in this documentation.

B. Cleaned Data Files

The analysis relies on the following cleaned datasets:

- **A_D_cleaned.csv:** Annual recreation visits and park attributes for Yellowstone and North Cascades, derived from the national annual visitation and park-metadata files.
- **BC_merged_cleaned_monthly_Yellowstone.csv:** Monthly Yellowstone visitation joined with Google Trends indices and metadata.
- **BC_merged_cleaned_monthly_NOCA.csv:** Monthly North Cascades visitation joined with Google Trends indices and metadata.
- **E_airbnb_cleaned.csv:** Airbnb listings for the Bozeman_Yellowstone and Seattle_NorthCascades regions after spatial, price, and minimum-night filtering and removal of missing key fields.

These files are the exact sources used to produce all tables, models, and visuals in the documentation.

C. Figure Generation (Matplotlib/Seaborn)

All charts in Section V were generated programmatically using Matplotlib and Seaborn from within the notebook using Python. Key examples include:

- **Annual and Monthly Visitation Plots:** line plots of RecreationVisits over Year and mean visits by Month.
- **Correlation Scatterplots:** RecreationVisits vs. trend_index for each park with fitted regression lines.
- **Airbnb Price Distributions:** boxplots and histograms of price by region and by room_type.
- **Time-Series Diagnostics:** residuals over time, residuals vs. predictions, residual histogram, and ACF plots for the Yellowstone time-series model.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error
from statsmodels.graphics.tsaplots import plot_acf

sns.set_style("whitegrid")

# 1. Load Yellowstone monthly data
yell_df = pd.read_csv("BC_merged_cleaned_monthly_Yellowstone.csv")

# Create a proper datetime column from Year and Month
yell_df["date"] = pd.to_datetime({
    "year": yell_df["Year"],
    "month": yell_df["Month"],
    "day": 1
})

# Sort by date just to be safe
yell_df = yell_df.sort_values("date").reset_index(drop=True)

# 2. Create lag features for time-series model
yell_df["visits_lag_1"] = yell_df["RecreationVisits"].shift(1)
yell_df["visits_lag_12"] = yell_df["RecreationVisits"].shift(12)

# Drop rows where lags OR trend_index are not available
yell_model_df = yell_df.dropna(subset=["visits_lag_1",
                                       "visits_lag_12",
                                       "trend_index"]).copy()

print(f"Rows after dropping NaNs: {len(yell_model_df)}")

# 3. Build feature matrix X and target y
month_dummies = pd.get_dummies(yell_model_df["Month"], prefix="month", drop_first=True)

feature_cols_basic = ["trend_index", "Year", "visits_lag_1", "visits_lag_12"]
X = pd.concat([yell_model_df[feature_cols_basic], month_dummies], axis=1)
y = yell_model_df["RecreationVisits"].values

# Extra safety: ensure no NaNs in X
nan_rows = X.isna().any(axis=1)
print(f"Rows with NaNs in X: {nan_rows.sum()}")
if nan_rows.sum() > 0:
    X = X[~nan_rows]
    y = y[~nan_rows]
    yell_model_df = yell_model_df[~nan_rows]

# 4. Train/test split (time-based)
train_mask = yell_model_df["Year"] <= 2019
test_mask = yell_model_df["Year"] >= 2020

X_train, X_test = X[train_mask], X[test_mask]
y_train, y_test = y[train_mask], y[test_mask]

dates_test = yell_model_df.loc[test_mask, "date"]

print(f"Train samples: {X_train.shape[0]}, Test samples: {X_test.shape[0]}")

# 5. Fit linear regression model
model_ts = LinearRegression()
model_ts.fit(X_train, y_train)

# 6. Predict on test set and compute metrics
y_pred_test = model_ts.predict(X_test)

r2_train = model_ts.score(X_train, y_train)
r2_test = r2_score(y_test, y_pred_test)
rmse_train = np.sqrt(mean_squared_error(y_train, model_ts.predict(X_train)))
rmse_test = np.sqrt(mean_squared_error(y_test, y_pred_test))
mae_test = mean_absolute_error(y_test, y_pred_test)

print("\n=== Yellowstone Time-Series Model (Model 4) ===")
print(f"Train R^2: {r2_train:0.3f}")
print(f"Test  R^2: {r2_test:0.3f}")
print(f"Train RMSE: {rmse_train:,.0f} visits")
print(f"Test RMSE: {rmse_test:,.0f} visits")
print(f"Test  MAE: {mae_test:,.0f} visits")

# 7. Build a DataFrame for diagnostics
diag_df = pd.DataFrame({
    "date": dates_test.values,
    "RecreationVisits": y_test,
    "y_pred_ts": y_pred_test
})
diag_df["residuals"] = diag_df["RecreationVisits"] - diag_df["y_pred_ts"]

# 8. Create residuals & diagnostics figure (4 panels)
fig, axes = plt.subplots(2, 2, figsize=(12, 8))

# (a) Residuals over time
axes[0, 0].plot(diag_df["date"], diag_df["residuals"], marker="o", linestyle="-")
axes[0, 0].axhline(0, color="red", linestyle="--", linewidth=1)
axes[0, 0].set_title("Residuals over Time (Test Period)")
axes[0, 0].set_xlabel("Date")
axes[0, 0].set_ylabel("Residuals")

# (b) Residuals vs. predicted
axes[0, 1].scatter(diag_df["y_pred_ts"], diag_df["residuals"], alpha=0.7)
axes[0, 1].axhline(0, color="red", linestyle="--", linewidth=1)
axes[0, 1].set_title("Residuals vs. Predicted Visits (Test Period)")
axes[0, 1].set_xlabel("Predicted Visits")
axes[0, 1].set_ylabel("Residuals")

# (c) Histogram of residuals
sns.histplot(diag_df["residuals"], bins=20, kde=True, ax=axes[1, 0])
axes[1, 0].axvline(0, color="red", linestyle="--", linewidth=1)
axes[1, 0].set_title("Distribution of Residuals (Test Period)")
axes[1, 0].set_xlabel("Residuals")
axes[1, 0].set_ylabel("Count")

# (d) ACF of residuals
plot_acf(diag_df["residuals"], lags=24, ax=axes[1, 1])
axes[1, 1].set_title("ACF of Residuals (Test Period)")

plt.tight_layout()
plt.savefig("Yellowstone_Time_Series_Model_Residuals_and_Diagnostics.png",
            dpi=300, bbox_inches="tight")
plt.show()
```

D. Tools and Environment

- Programming language: Python 3
- Environment: Google Colaboratory
- Core libraries: pandas, NumPy, scikit-learn, Matplotlib, Seaborn, and statsmodels

X. Student Copyright Checklist

STUDENT COPYRIGHT CHECKLIST *(for students to complete and advisors to verify)*

STUDENT: Answer question 1 below.

1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☒ NO
   If NO, go to question 2.
   If YES, is the music and/or sound copyrighted? ☐ YES ☐ NO
   If YES, move to question 1A. If NO, move to question 1B.
   1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.
   1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, John Kevin Finney (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

2) Does your solution to the competitive event integrate any graphics/videos? ☒ YES ☐ NO
   If NO, go to question 3.
   If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☒ NO
   If YES, move to question 2A. If NO, move to question 2B.
   2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.
   2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

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I, John Kevin Finney (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

3) Does your solution to the competitive event use another's thoughts or research? ☒ YES ☐ NO
   If NO, this is the end of the checklist.
   If YES, have you properly cited other's thoughts or research in your documentation? ☒ YES ☐ NO

   CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

   I, John Kevin Finney (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Initials: R.K.

Chapter Advisor Signature: John K Finney

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STUDENT: Answer question 1 below.

1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☑ NO
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   If YES, is the music and/or sound copyrighted? ☐ YES ☐ NO
   If YES, move to question 1A. If NO, move to question 1B.
   1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.
   1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

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STUDENT: Answer question 2 below.

2) Does your solution to the competitive event integrate any graphics/videos? ☑ YES ☐ NO
   If NO, go to question 3.
   If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☑ NO
   If YES, move to question 2A. If NO, move to question 2B.
   2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.
   2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

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STUDENT: Answer question 3 below.

3) Does your solution to the competitive event use another's thoughts or research? ☑ YES ☐ NO
   If NO, this is the end of the checklist.
   If YES, have you properly cited other's thoughts or research in your documentation? ☑ YES ☐ NO

   CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

   I, John Kevin Finney (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Initials: S.I.

Chapter Advisor Signature: John K Finney